

CSE331

AUTAMATA AND COMPUTABILITY

Lecture 11

Spring 2026

Today's learning goals

Sipser Ch 1.4

- Explain the limits of the class of regular languages
- Justify why the Pumping Lemma is true
- Apply the Pumping Lemma in proofs of non-regularity
- Identify some nonregular languages

All roads lead to ... regular sets?

Are there any languages over $\{0,1\}$ that are not regular?

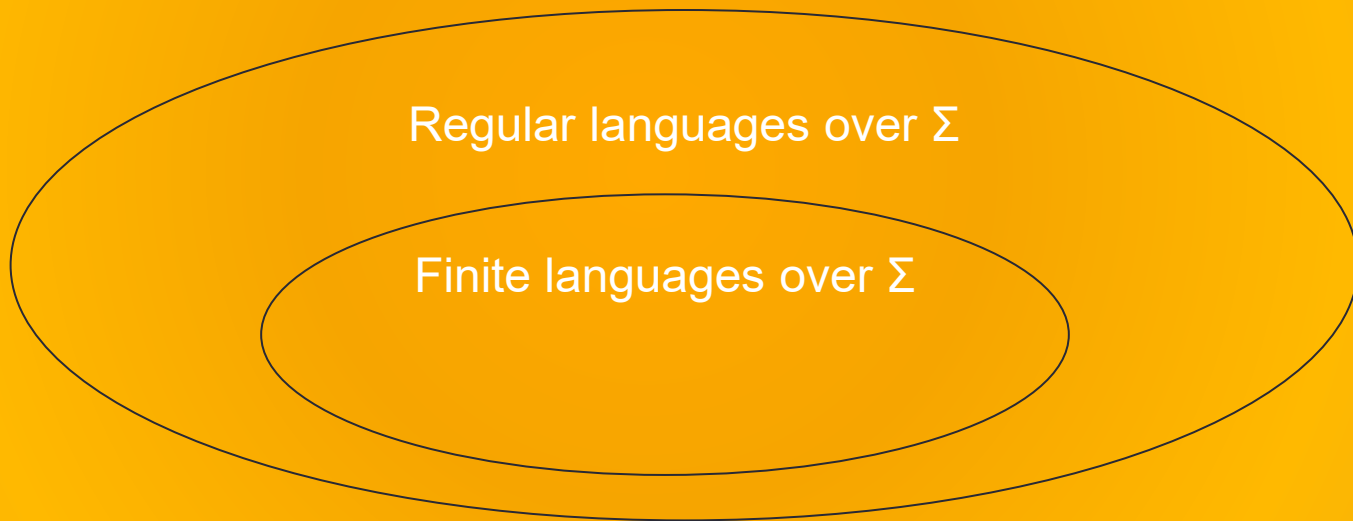
- A. Yes: a language that is recognized by an NFA but not any DFA.
- B. Yes: there is some infinite language of strings over $\{0,1\}$ that is not described by any regular expression.
- C. No: all languages over $\{0,1\}$ are regular because that's what it means to be a language.
- D. No: for each set of strings over $\{0,1\}$, some DFA recognizes that set.
- E. I don't know.

Birds' eye view

All languages over Σ

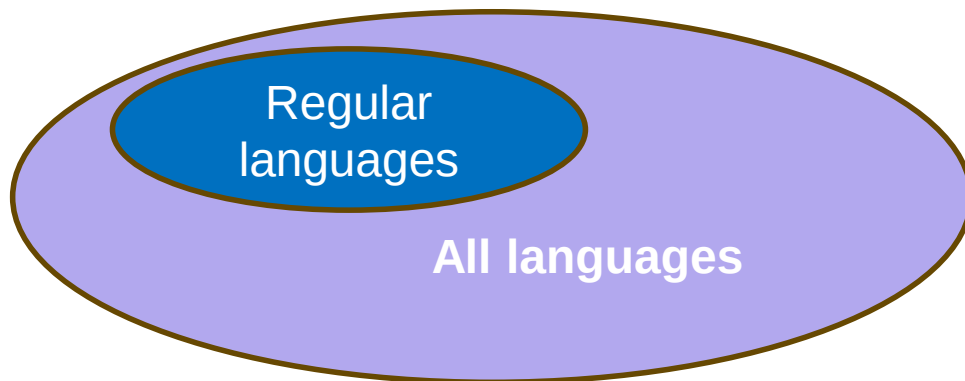
Regular languages over Σ

Finite languages over Σ



Counting

- There are more **languages** than **regular languages**
- Conclusion: there must be non-regular languages
- But how can we identify them???



Proving nonregularity

How can we prove that a set is non-regular?

- A. Try to design a DFA that recognizes it and, if the first few attempts don't work, conclude there is none that does.
- B. Prove that it's a strict subset of some regular set.
- C. Prove that it's the union of two regular sets.
- D. Prove that its complement is not regular.
- E. I don't know.

Where we stand

- There exist non-regular languages
- If we know that some sets are not regular, we can conclude others are **also** not regular **judiciously reasoning using closure properties of class of regular languages.**
- No example of a specific non-regular set **... yet.**

Bounds on DFA

- in DFA, memory = states
- Automata can only "remember" ...
 - ...finitely far in the past
 - ...finitely much information
- If a computation path visits the same state more than once, the machine can't tell the difference between the first time and future times it visited that state.

Example!

$$\{ 0^n 1^n \mid n \geq 0 \}$$

What are some strings in this set?

What are some strings not in this set?

*Compare to $L(0^*1^*)$*

Design a DFA? NFA?

Example!

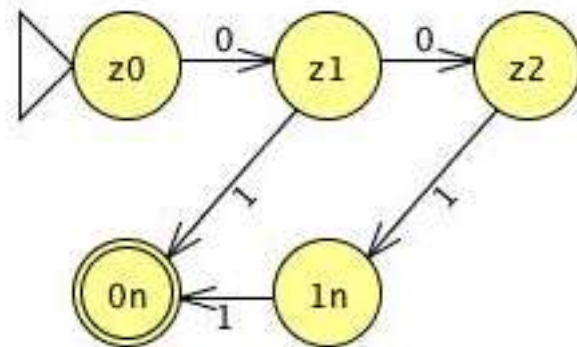
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What are some strings in this set?

What are some strings not in this set?

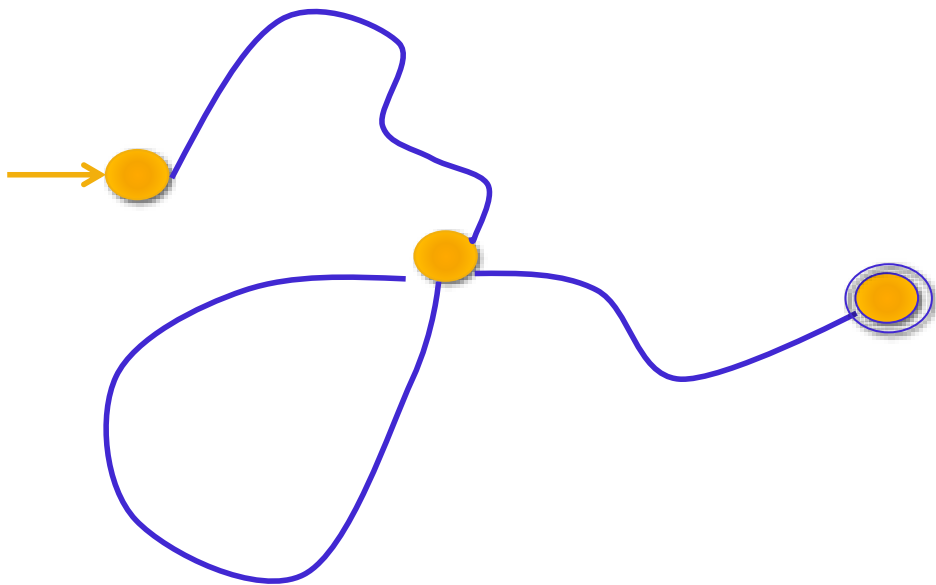
*Compare to $L(0^*1^*)$*

Design a DFA? NFA?



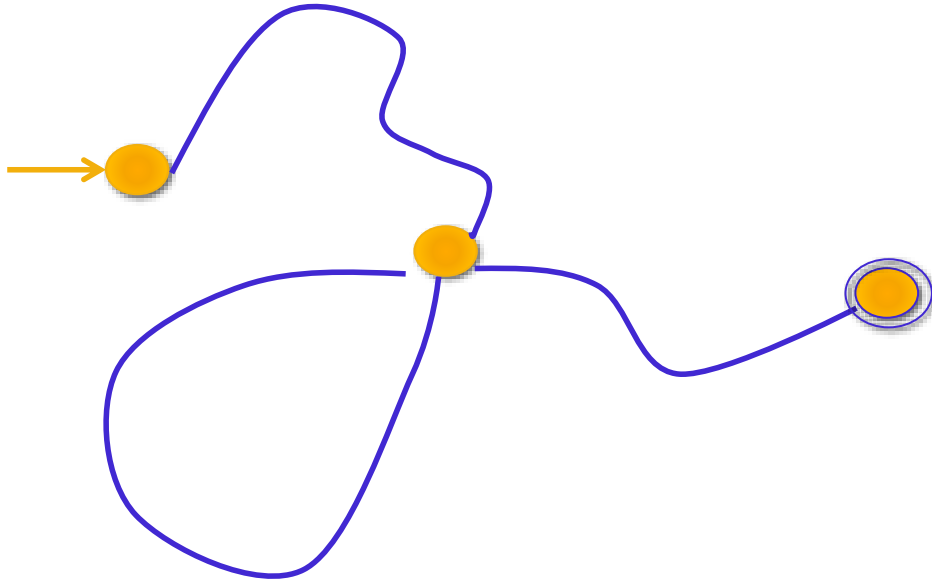
Pumping

- Focus on computation path through DFA



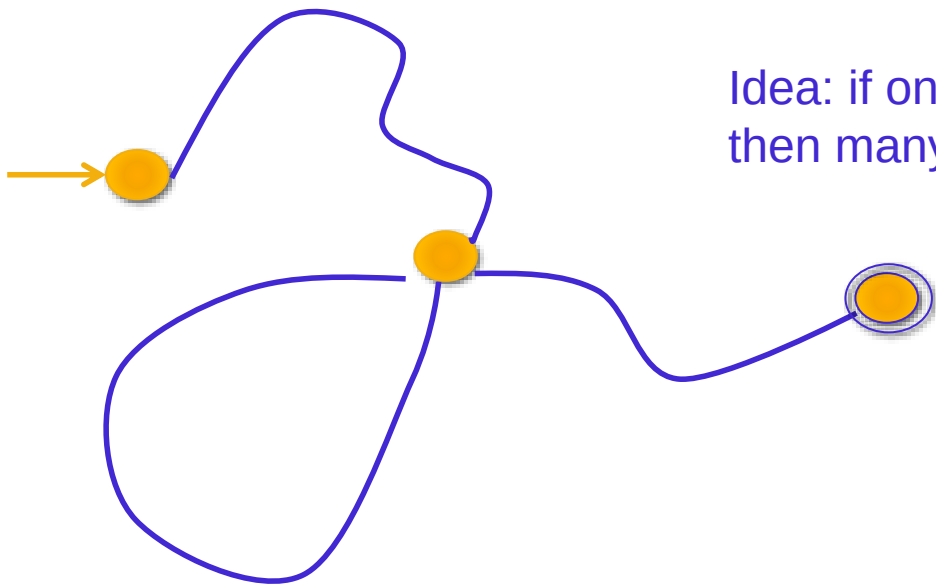
Pumping

- Focus on computation path through DFA



Pumping

- Focus on computation path through DFA



Idea: if one **long** string is accepted, then many other strings have to be accepted too

Pumping Lemma

Sipser p. 78 Theorem 1.70

If A is a regular language, then there is a number p (*the pumping length*) where, if s is any string in A of length at least p , then s may be divided into three pieces, $s = x y z$ such that

- $|y| > 0$, and
- for each $i \geq 0$, $xy^iz \in A$,
- $|xy| \leq p$.

Pumping Lemma

Sipser p. 78 Theorem 1.70

If A is a regular language, then there is a number p (*the pumping length*) where, if s is any string in A of length at least p , then s may be divided into three pieces, $s = x y z$ such that

states in DFA
recognizing A

- $|y| > 0$, and
- for each $i \geq 0$, $xy^iz \in A$,
- $|xy| \leq p$.

Transition labels
along loop

Proof strategy

To prove that a language L is **not** regular

- Assume towards a contradiction that it is.
- Use Pumping Lemma to **give** p , a pumping length for L
- Show that p actually isn't a pumping length for L .
- $\rightarrow\leftarrow$
- Conclude that L is not regular.

Using the Pumping Lemma

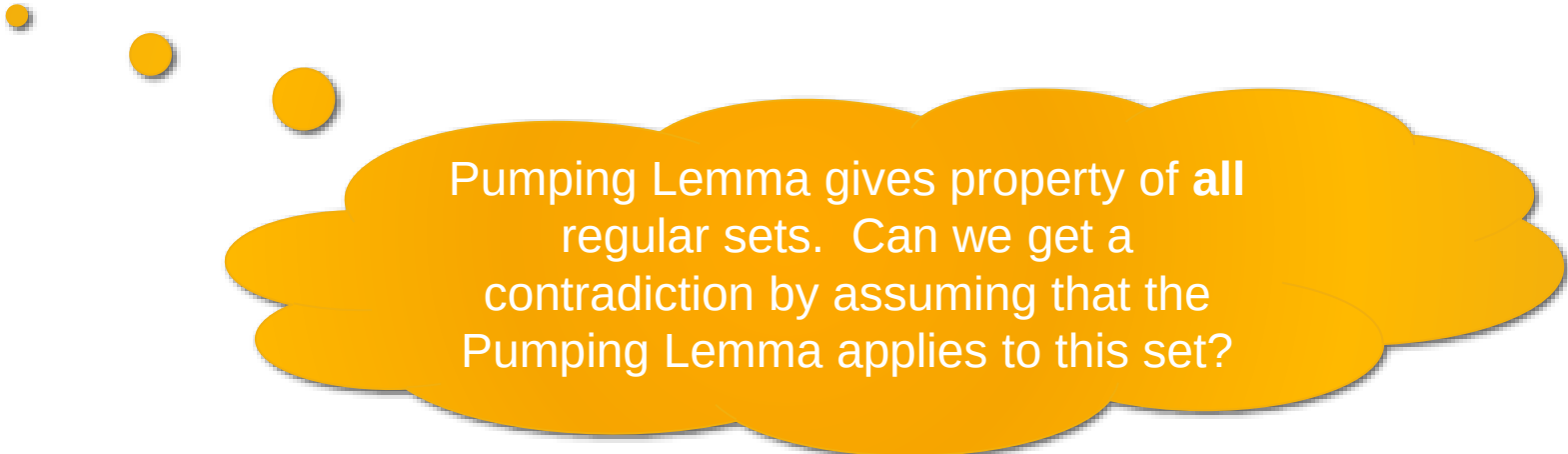
Claim: The set $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Using the Pumping Lemma

Claim: The set $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof:

Assume, **towards a contradiction**, that L is regular.



Pumping Lemma gives property of **all** regular sets. Can we get a contradiction by assuming that the Pumping Lemma applies to this set?

Using the Pumping Lemma

Claim: The set $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof:

Assume towards a contradiction L is regular.

So by Pumping Lemma, L has a pumping length, call it p .

FACT: p is a pumping length for L (by definition).

CLAIM: p is not a pumping length for L .

Conclude: contradiction!

Using the Pumping Lemma

Claim: The set $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof: ...In particular, this means that every string in L that is of length p or more can be "pumped".

Goal: pick a string s in L of length greater than or equal to p such that any division of s as $s = xyz$ with $|y| > 0$ and $|xy| \leq p$ gives some value $i \geq 0$ with $xy^i z$ not in L

So we have a contradiction, and L is not regular.

Using the Pumping Lemma

Claim: The set $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof: ...

Goal: pick a string s in L of length greater than or equal to p such that any division of s as $s = xyz$ with $|y| > 0$ and $|xy| \leq p$ gives some value $i \geq 0$ with $xy^i z$ not in L

Choose $s = 0^p 1^p$. Consider any $s = xyz$ with $|y| > 0$, $|xy| \leq p$.

Using the Pumping Lemma

Claim: The set $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof: ...

Goal: pick a string s in L of length greater than or equal to p such that any division of s as $s = xyz$ with

$|y| > 0$ and $|xy| \leq p$ gives some value $i \geq 0$ with $xy^i z$ not in L

Choose $s = 0^p 1^p$. Consider any $s = xyz$ with $|y| > 0$, $|xy| \leq p$.

Since $|xy| \leq p$, $x = 0^k$, $y = 0^m$, $z = 0^r 1^p$ with $k+m+r = p$, $m > 0$.

Using the Pumping Lemma

Claim: The set $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof: ...

Goal: pick a string s in L of length greater than or equal to p such that any division of s as $s = xyz$ with

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Picking $i=0$: $xy^i z = xz = 0^k 0^r 1^p = 0^{k+r} 1^p$, not in L !

Using the Pumping Lemma

Claim: The set $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof: ...

Goal: pick a string s in L of length greater than or equal to p such that any division of s as $s = xyz$ with $|y| > 0$ and $|xy| \leq p$ gives some value $i \geq 0$ with $xy^i z$ not in L

Choose $s = 0^p 1^p$. Consider any $s = xyz$ with $|y| > 0$, $|xy| \leq p$.

Since $|xy| \leq p$, $x = 0^k$, $y = 0^m$, $z = 0^r 1^p$ with $m+n+r = p$, $m > 0$.

Picking $i=0$: $xy^i z = xz = 0^k 0^r 1^p = 0^{k+r} 1^p$, not in L ! This is a contradiction with the Pumping Lemma applied to L , so L must not be regular.



The pumping lemma game

- The logic in applying the pumping lemma might seem confusing
- Can think of it as a game between **you (the prover)** and an **adversary (claiming that the language is regular)**
- Your goal is to prove that the adversary is wrong, and the language **is not regular**

Pumping lemma as a game

- Two players: you (prover) and adversary



p (pumping length)



$w \in L$ with $|w| \geq p$



$w = xyz$ with $|xy| \leq p, |y| \geq 1$



$i \geq 0$ such that $xy^iz \notin L$



Pumping lemma as a game

- Two players: you (prover) and adversary

Your choices



p (pumping length)

$w \in L$ with $|w| \geq p$

$w = xyz$ with $|xy| \leq p, |y| \geq 1$

$i \geq 0$ such that $xy^iz \notin L$

Not your choices



Another example

Claim: The set $\{a^n b^m a^n \mid m, n \geq 0\}$ is not regular.

Proof: ... **Consider the string $s = \dots$**

You must pick s carefully: we want $|s| \geq p$ and s in L . Now we will prove a contradiction with the statement " s can be pumped"

Which choices of s cannot be used to complete the proof?

A. $s = a^p b^p$ **B.** $s = ab^p a$ **C.** $s = a^p b^p a^p$ **D.** $s = a^p b a^p$

E. None of the above (all of these choices work).

Another example

Claim: The set $\{a^n b^m a^n \mid m, n \geq 0\}$ is not regular.

Proof: ... **Consider the string $s = \dots$**

You must pick s carefully: we want $|s| \geq p$ and s in L . Now we will prove a contradiction with the statement " s can be pumped"

Consider an arbitrary choice of x, y, z such that $s = xyz$, $|y| > 0$, $|xy| \leq p$. **This means that...** What properties are guaranteed about x, y, z ?

Consider $i=...$ In this case, $xy^iz = \dots$, which is not in L , a contradiction with the Pumping Lemma applying to L and so L is not regular.

And another

Claim: The set $\{w w^R \mid w \text{ is a string over } \{0,1\}\}$ is not regular.

Proof: ...**Consider the string $s = \dots$**

You must pick s carefully: we want $|s| \geq p$ and s in L . Now we will prove a contradiction with the statement " s can be pumped" **Consider $i = \dots$**

Which s and i let us complete the proof?

- A.** $s = 0^p 0^p$, $i=2$ **B.** $s = 0110$, $i=0$ **C.** $s = 0^p 110^p$, $i=1$ **D.** $s = 1^p 001^p$, $i=3$
E. I don't know

How do we choose i ?

Claim: The set $\{0^j 1^k \mid j, k \geq 0 \text{ and } j \geq k\}$ is not regular.

Proof: ... **Consider the string $s = \dots$**

You must pick s carefully: we want $|s| \geq p$ and s in L . Now we will prove a contradiction with the statement " s can be pumped" **Consider $i = \dots$**

Which s and i let us complete the proof?

- A. $s = 0^p 1^p$, $i=2$ B. $s = 0^p 1^p$, $i=p$ C. $s = 0^p 1^p$, $i=1$ D. $s = 0^p 1^p$, $i=0$
E. I don't know

Regular sets: not the end of the story

- Many **nice / simple / important** sets are not regular
- Limitation of the finite-state automaton model
 - Can't "count"
 - Can only remember finitely far into the past
 - Can't backtrack
 - Must make decisions in "real-time"
- We know computers are more powerful than this model...

Which conditions should we relax?

The next model of computation

- **Idea:** allow *some* memory of unbounded size
- **How?**
 - Generalization of regular expressions → **Context-free grammars**
 - Generalization for DFA → **Pushdown Automata**